

Summer School · EIV 2026

GNNs & LLMs: Reasoning Beyond Text

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Guest at Escuela Colombiana de Ingeniería Julio Garavito
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5 days · 24 hrs · In-Person



SECTION 0

Module 3 Recap

Module 3 Recap

III.1. Un grafo G se define formalmente como... (una opción correcta)

- a. Un conjunto de N nodos (vertices) y un conjunto de edges (aristas): $G = \langle N, E \rangle$
- b. Una tabla donde cada fila N es una entidad independiente $G = \langle N \rangle$
- c. Una secuencia ordenada de tokens con posiciones fijas
- d. Una matriz de features sin información de conectividad

Module 3 Recap

III.2. ¿Por qué los grafos preservan información que las tablas pierden? (una opción correcta)

- a. Porque la estructura (las relaciones/conexiones entre entidades) es en sí misma una señal.
- b. Porque los grafos almacenan más features por entidad que una tabla
- c. Porque las tablas no pueden almacenar datos categóricos
- d. Porque los grafos comprimen los datos y ocupan menos memoria

Module 3 Recap

III.3. El *Node Feature Matrix* X de un grafo con N nodos y D features por nodo tiene dimensión... (una opción correcta)

- a. $N \times D$
- b. $N \times N$
- c. $D \times D$
- d. $2 \times E$, donde E es el número de edges

Module 3 Recap

III.4. ¿Por qué el deep learning convencional (MLPs, CNNs) es inadecuado para procesar grafos directamente? (una opción correcta)

- a. Porque asume entradas de estructura y tamaño fijos, mientras que los grafos tienen tamaño variable y no existe una noción fija de localidad ni de sliding window
- b. Porque los grafos contienen demasiados datos para entrenar redes neuronales
- c. Porque los MLPs y las CNNs no pueden procesar vectores de features numéricos
- d. Porque los grafos solo admiten métodos no basados en gradientes

Module 3 Recap

III.5. ¿Cuáles son los tres niveles de predicción típicos en graph representation learning? (una opción correcta)

- a. Node-level, link-level y graph-level
- b. Token-level, sentence-level y document-level
- c. Pixel-level, patch-level e image-level
- d. Training-level, validation-level y test-level

Module 3 Recap

III.6. En el esquema de message passing de una capa GNN, el orden correcto de las operaciones para actualizar un nodo i es... (una opción correcta)

- a. Message $\rightarrow$ Aggregate $\rightarrow$ Update
- b. Update $\rightarrow$ Aggregate $\rightarrow$ Message
- c. Aggregate $\rightarrow$ Message $\rightarrow$ Update
- d. Message $\rightarrow$ Update $\rightarrow$ Aggregate, aplicado solo a los nodos sin vecinos

Module 3 Recap

III.7. Sobre las funciones de message (m) y update (u) en una GNN, es correcto afirmar que... (una opción correcta)

- a. Son redes neuronales (p. ej. MLPs) cuyos parámetros se comparten entre todos los nodos del grafo
- b. Cada nodo del grafo aprende sus propias funciones m y u con parámetros independientes
- c. Son funciones fijas sin parámetros aprendibles, como el promedio simple
- d. Solo existen en grafos dirigidos

Module 3 Recap

III.8. ¿Cuál es la idea central del Graph Attention Network (GAT)? (una opción correcta)

- a. Aprender coeficientes de attention α_{ij} que ponderan de forma diferente la contribución de cada vecino durante la aggregation, lo cual (con multi-head) ayuda a manejar la heterofilia
- b. Asignar a todos los vecinos exactamente el mismo peso $1/|N(v)|$
- c. Reemplazar el message passing por una matriz de adyacencia aprendida desde cero
- d. Eliminar los edges con pesos bajos antes de entrenar

Module 3 Recap

III.9. Al apilar K capas de GNN, el receptive field de cada nodo... (una opción correcta)

- a. Se expande a su vecindad de K hops: cada capa adicional permite incorporar información de vecinos un hop más lejos
- b. Permanece fijo en los vecinos directos (1 hop), sin importar K
- c. Se reduce, porque cada capa filtra vecinos
- d. Cubre siempre el grafo completo desde la primera capa

Course Overview

I · 5-6 hrs

ML Foundations

ML best practices. Train first ML model.

II · 5-6 hrs

LLMs + RAG

Fundamentals of LLMs and first RAG system.

III · 5-6 hrs

GNNs

Encode relational structure.

IV · 5-6 hrs

LLMs + GNNs

Combine all three end-to-end

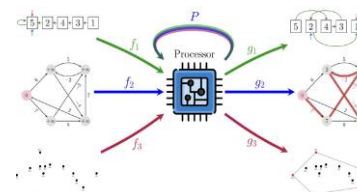
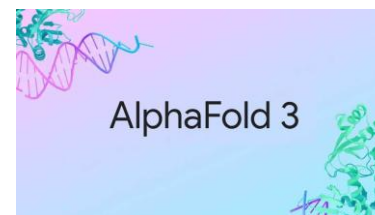
V · 3-4 hrs

Wrap-up

Exam and extra topic

Why?

	Write me an email/code	"Explain X to me"	Turn off lights	Algebra and algorithms	Fold this protein	Solve this NP-Hard problem
LLMs						
GNNs						



Gemini

Scale

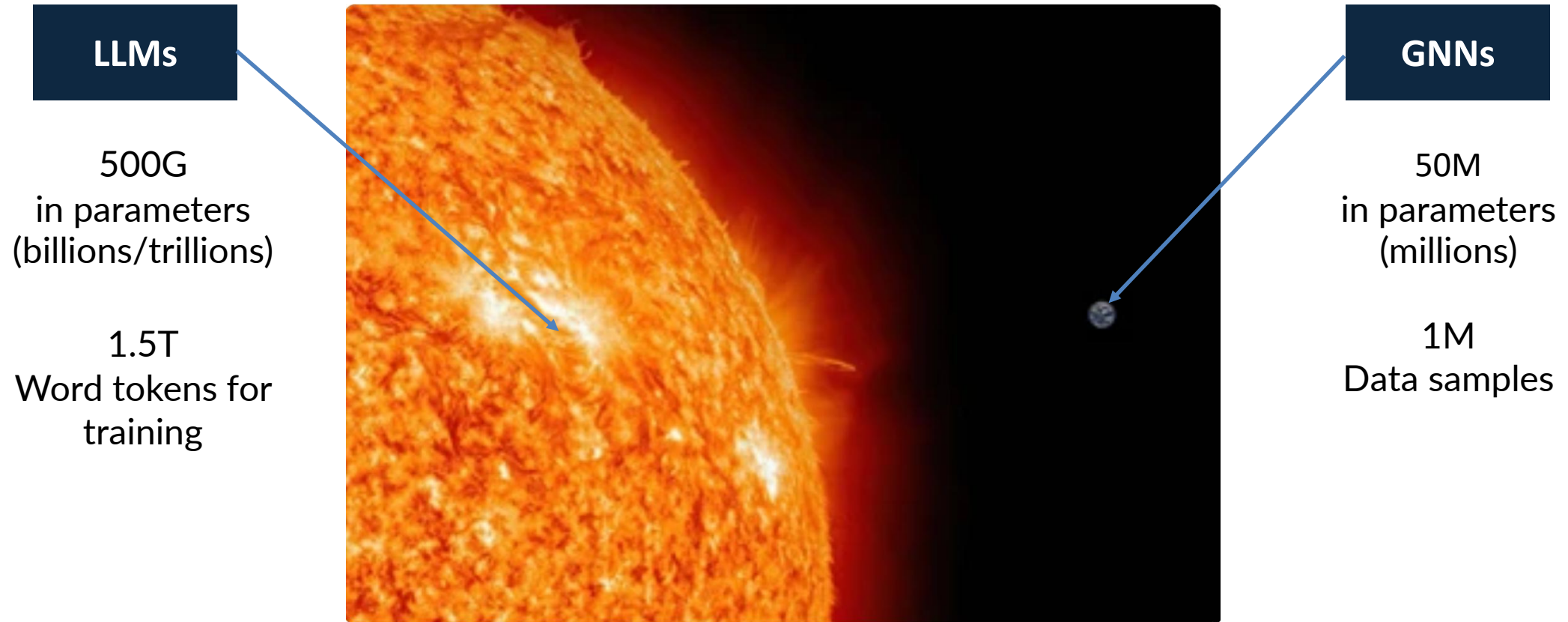


Image idea from Assoc. Prof. Xavier Bresson, LoG conference Keynote 2024.

From Module III to Module IV

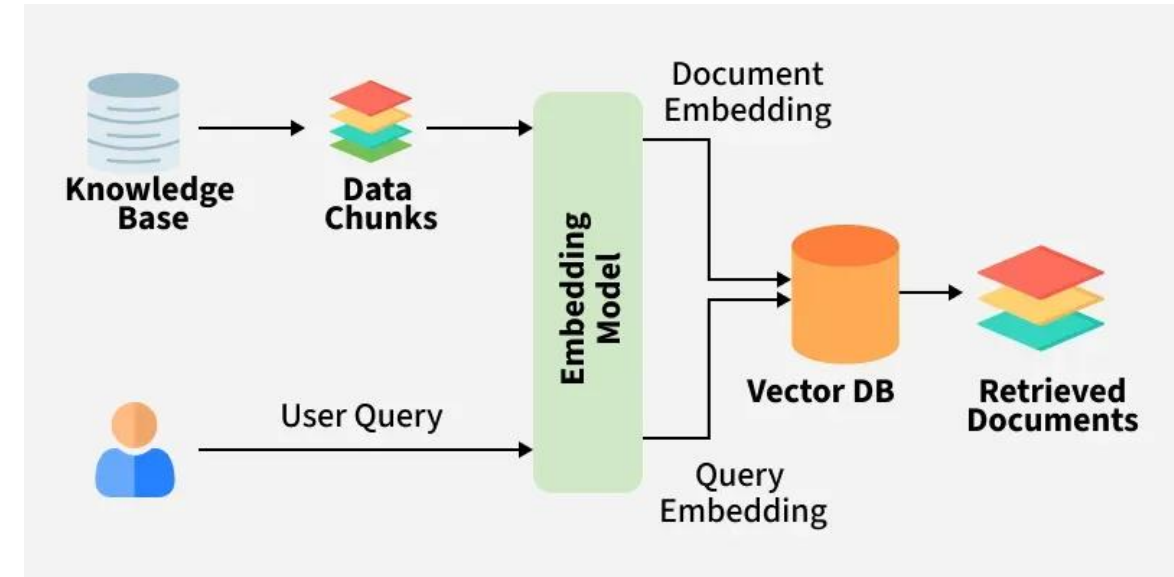
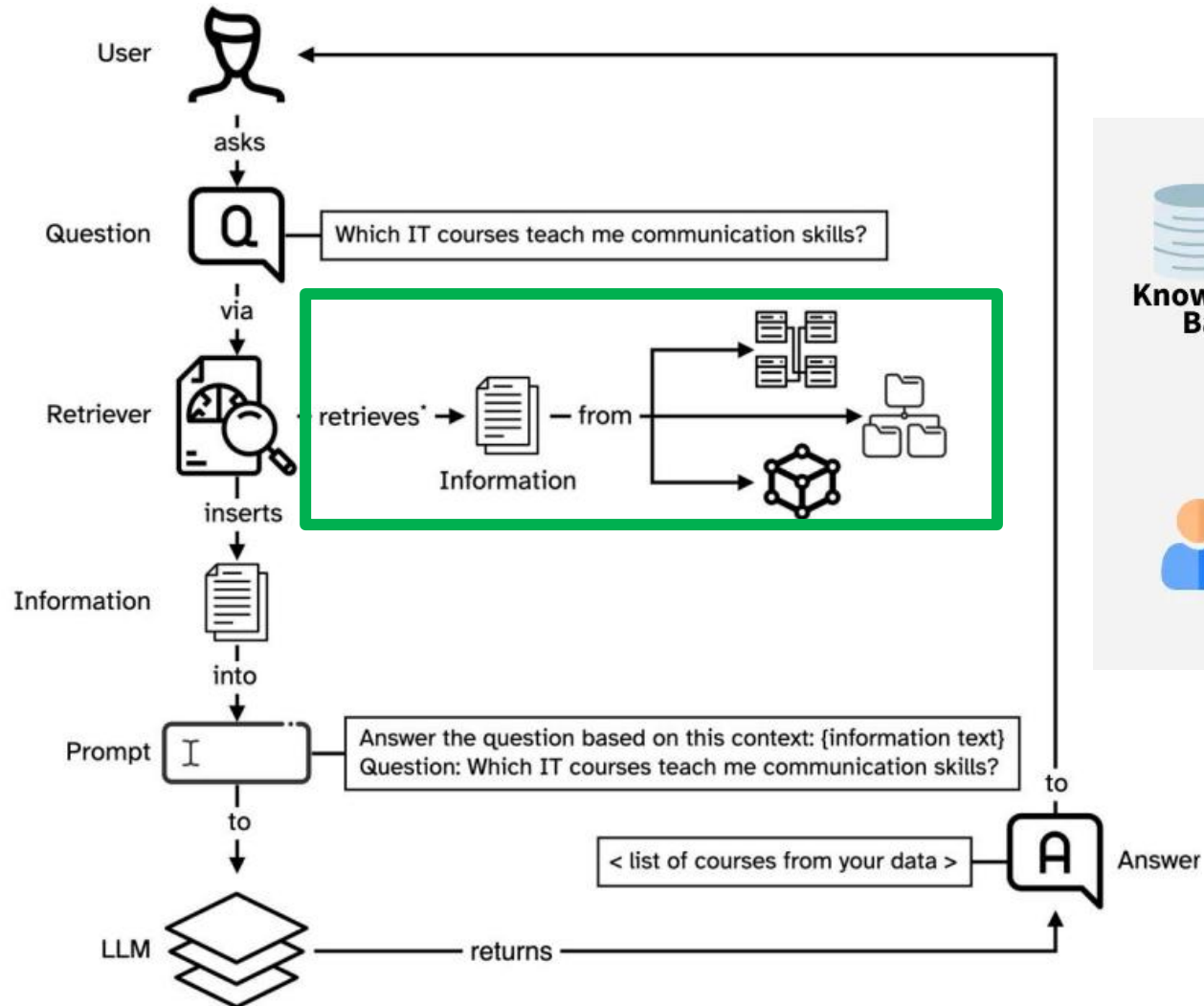
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- The **LLM** reasons over text — but never sees the graph
- The **GNN** reasons over structure — but can't speak language
- RAG bridges to documents — but the edges are **lost** the moment text is chunked

Make the graph's structure available to the language model itself — one system that reads and reasons.

The roadmap: text-attributed graphs → GraphRAG → G-Retriever → an LLM that reasons over relationships.

RAG – Recap

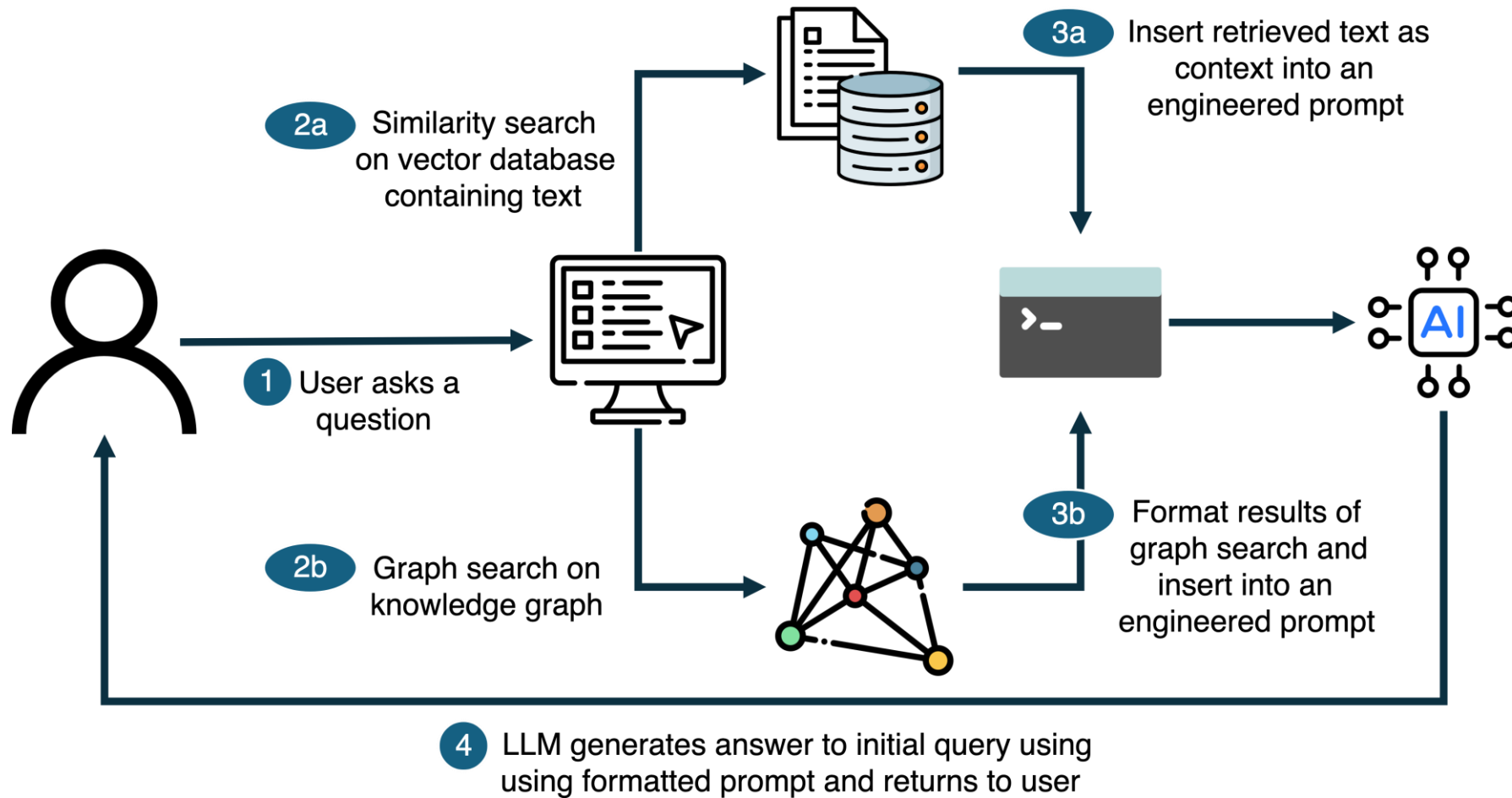


Significantly improves responses on your dataset

No relational data support!

What is the relationship between retrieved chunks?

GraphRAG



GraphRAG

WE DON'T HAVE THE GRAPH

Traditional GraphRAG proposed by Microsoft in 2024.

Use an LLM to convert non-structured data into a graph.



WE HAVE THE GRAPH

Directly use the Graph to extract the structural information from the DB



SECTION 1

Text-Attributed Graphs

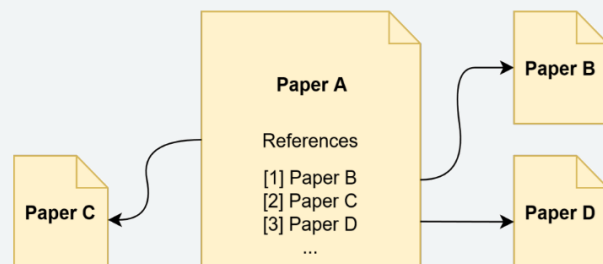
What is a Text-Attributed Graph?

$G = (N, E, T)$ a graph whose nodes (and often edges) each carry a piece of **natural-language text** — T : node $\rightarrow$ string

Citation networks

Nodes = papers (title + abstract).

Edges = citations.

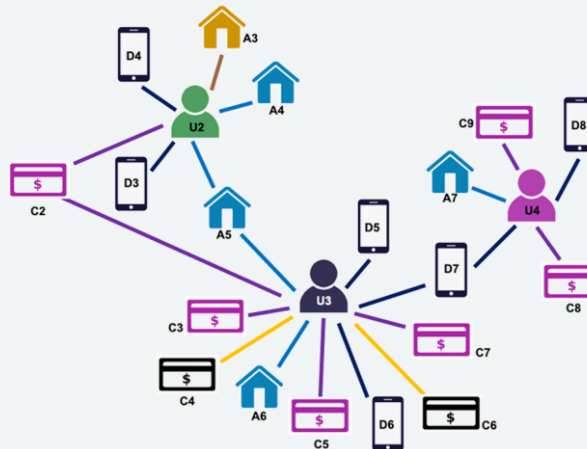


Cora · ogbn-arxiv

Product graphs

Nodes = items (description + reviews).

Edges = co-purchase.

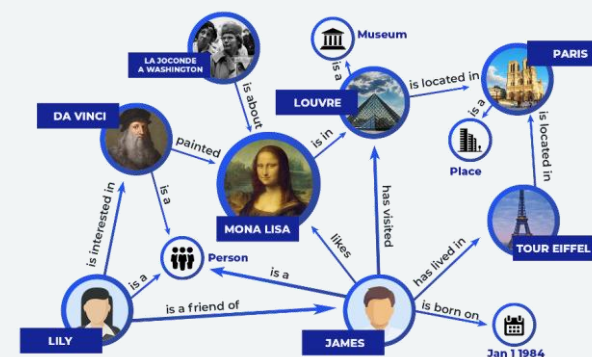


Amazon · ogbn-products

Knowledge graphs

Nodes = entities,

edges = named relations — both in language.



Wikidata · enterprise KG

Knowledge Graph

Way of organizing information as a **network of entities** and the **relationships** between them, rather than as rows in tables or plain documents.

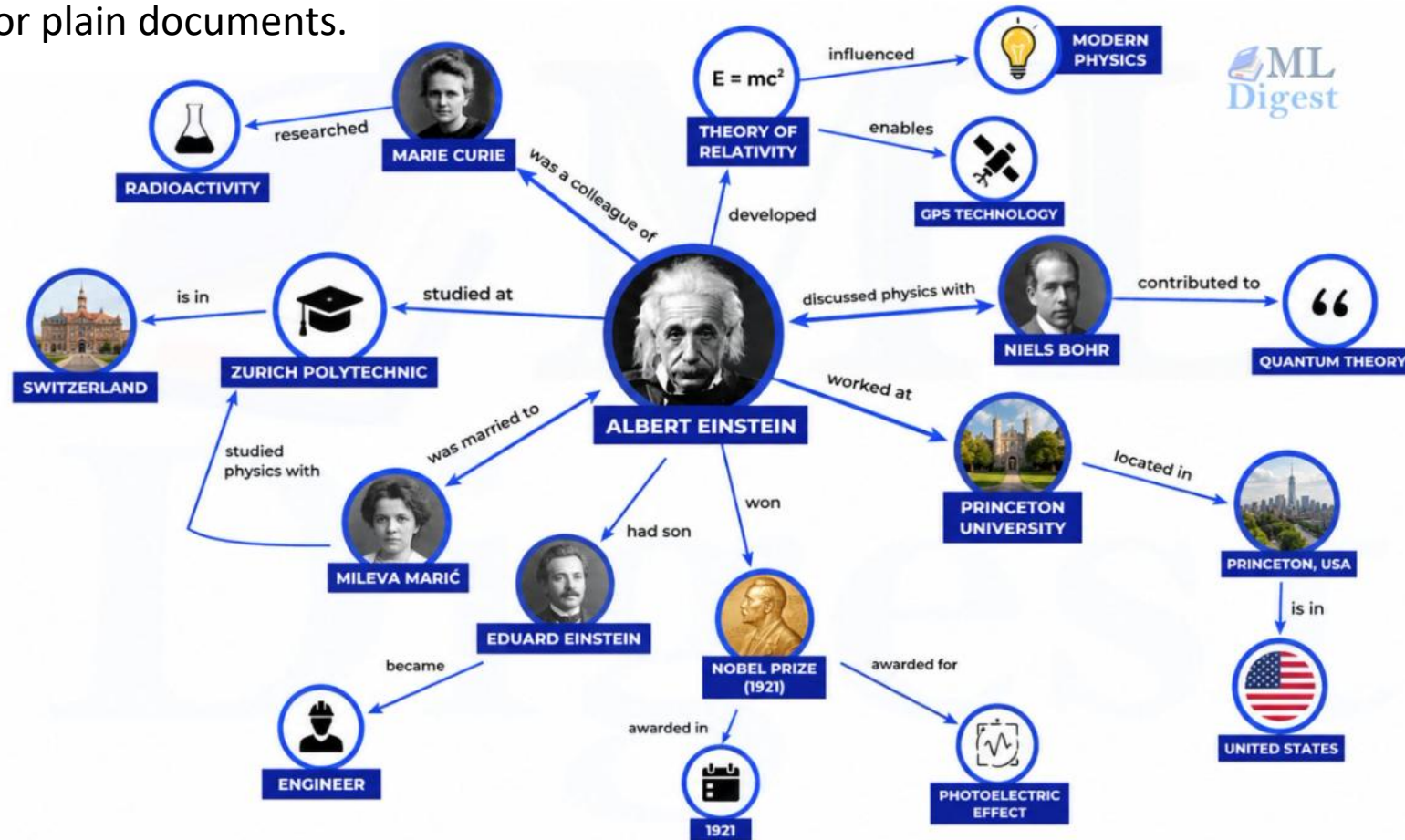


Image credit: ML digest. What Are Knowledge Graphs? A Comprehensive Guide to Connected Data.

Converting Text to Graph

Table 1:

employee_id	name	department_id
101	Alice	5

Table 2:

dep_id	name
5	Data Science



Two Signals in One Object

SEMANTICS

What each node says

The meaning inside the node's text — entities, claims, nuance.

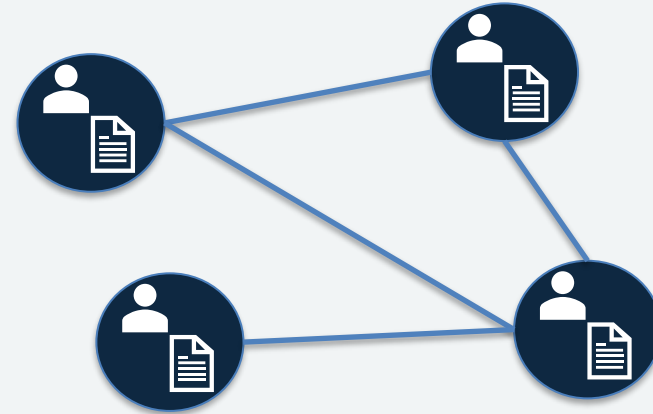


Captured by → [Language models](#)

STRUCTURE

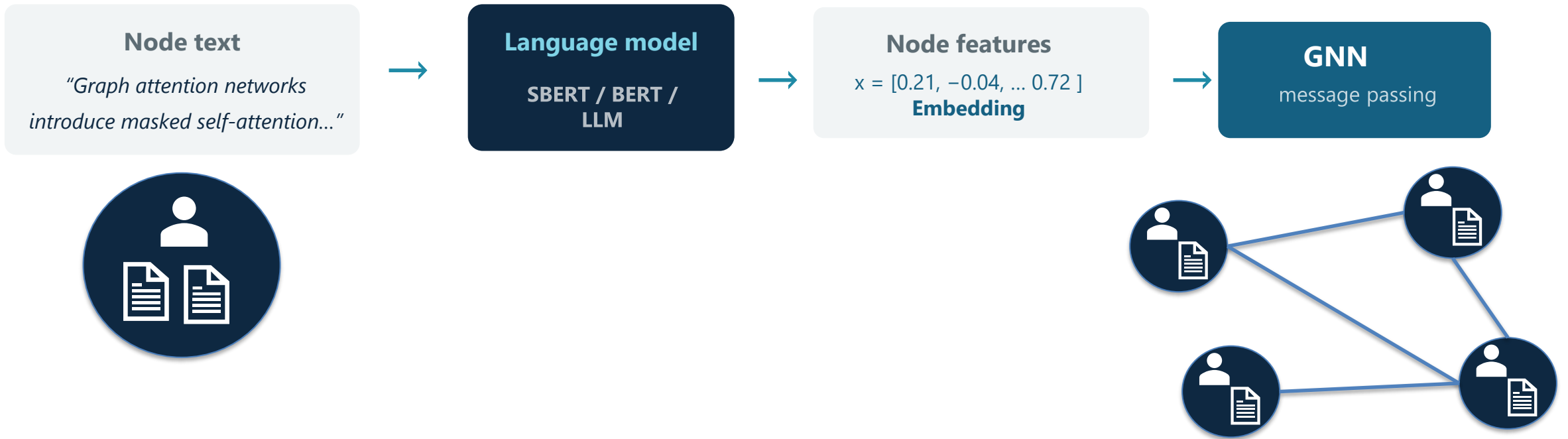
How nodes connect

The neighbourhood — who relates to whom, and how far apart.



Captured by → [Graph neural networks](#)

Encoding Node Text into Features



SECTION 2

Traditional GraphRAG

Remember: plain RAG



Great at

Single-fact lookup — “What is the GAT attention coefficient?” One chunk has the answer.

Breaks on

Multi-hop & relationships — chunks are retrieved **independently** ; there are no edges to follow.

The GraphRAG Idea

PLAIN RAG INDEX

☐ chunk 17

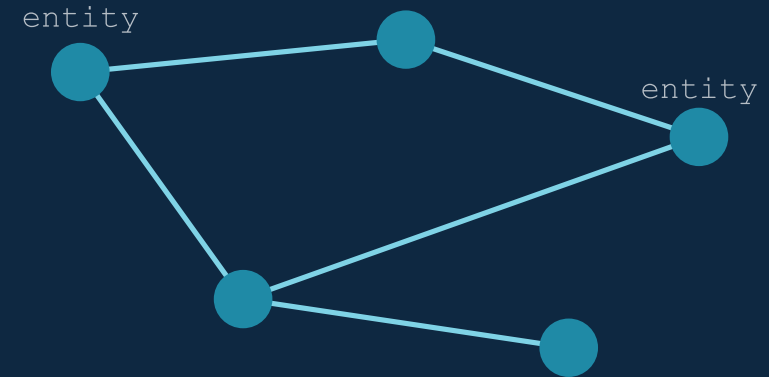
☐ chunk 42

☐ chunk 88

A flat list — no connections.



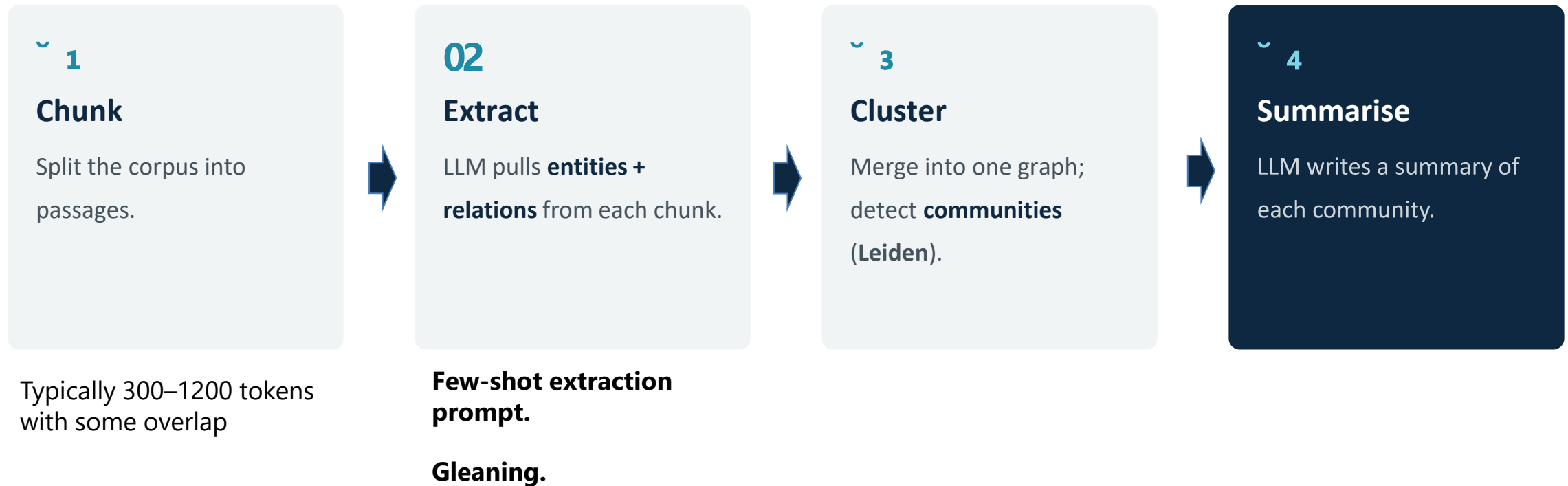
GRAPHRAG INDEX



Entities + relations — retrieval can traverse.

Build a knowledge graph from the corpus, then retrieve over its structure — not a flat list of passages.

Building the graph — the LLM “does” it



Microsoft GraphRAG (Edge et al., 2024). Next: each step in detail.

Step 1–2: the LLM turns text into triples

ONE CHUNK OF THE CORPUS

Ada Lovelace collaborated with Charles Babbage on the Analytical Engine, which Babbage designed in London.

EXTRACTED TRIPLES (subject — relation → object)

Ada Lovelace —collaborated_with→ Charles Babbage

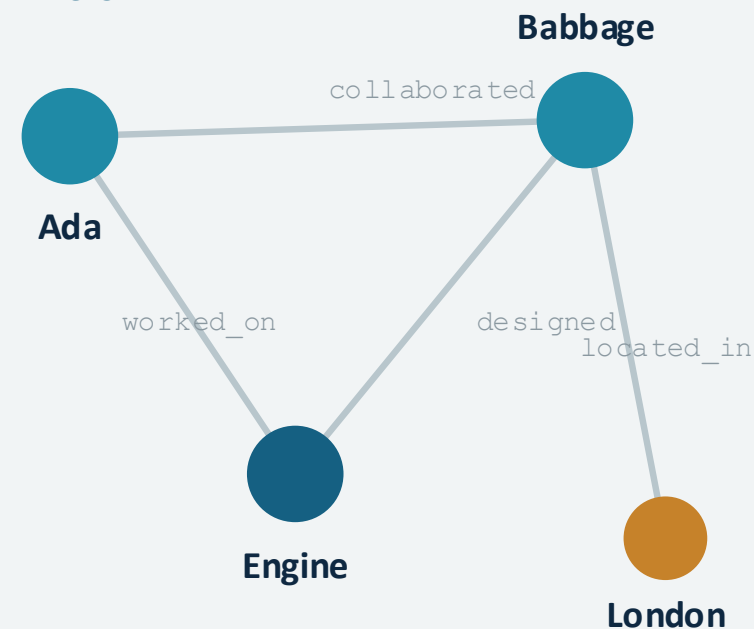
Ada Lovelace —worked_on→ Analytical Engine

Charles Babbage —designed→ Analytical Engine

Charles Babbage —located_in→ London

The relation is a sentence, not a predicate from a fixed vocabulary.

→ NODES + EDGES



Step 2: Extract through Few-Shot Prompting

*"Given a text document and a list of entity types, identify all entities of those types and all relationships among the identified entities. For each entity **extract: entity_name, entity_type, entity_description** (comprehensive description of the entity's attributes and activities). For **each pair of related entities extract: source_entity, target_entity, relationship_description** (explanation of why they are related), **relationship_strength** (an integer score). Output as a delimited list..."*



Descriptions instead of predicates

```
("entity" | NEOCHIP | ORGANIZATION | NeoChip is a chip manufacturer that announced record Q3 earnings driven by its Fusion processor line)
("entity" | SARAH LIN | PERSON | Sarah Lin is the CEO of NeoChip who credited the Fusion processors for strong earnings)
("relationship" | SARAH LIN | NEOCHIP | Sarah Lin is the chief executive officer of NeoChip | 9)
("relationship" | NEOCHIP | FUSION PROCESSOR | NeoChip designs and sells the Fusion processor line, its main revenue driver | 8)
```

Gleaning

Continuation

"MANY entities and relationships were missed in the last extraction. Add them below using the same format..."

Loop-check

"It appears some entities and relationships may have still been missed. Answer Y or N if there are still entities or relationships that need to be added..."

Step 2: Extract through Few-Shot Prompting – Result

-----Entities-----

id	entity	description
1	ACME CORP	Manufacturing firm specializing in industrial sensors...
2	GLOBEX	Logistics conglomerate headquartered in Rotterdam...
3	JANE DOE	CEO of Acme Corp since 2021, previously at...

-----Relationships-----

source	target	description	weight
ACME CORP	GLOBEX	Acme supplies sensor components to Globex	
		under a 2023 exclusivity agreement	8
JANE DOE	ACME	Jane Doe leads Acme Corp as chief executive	9

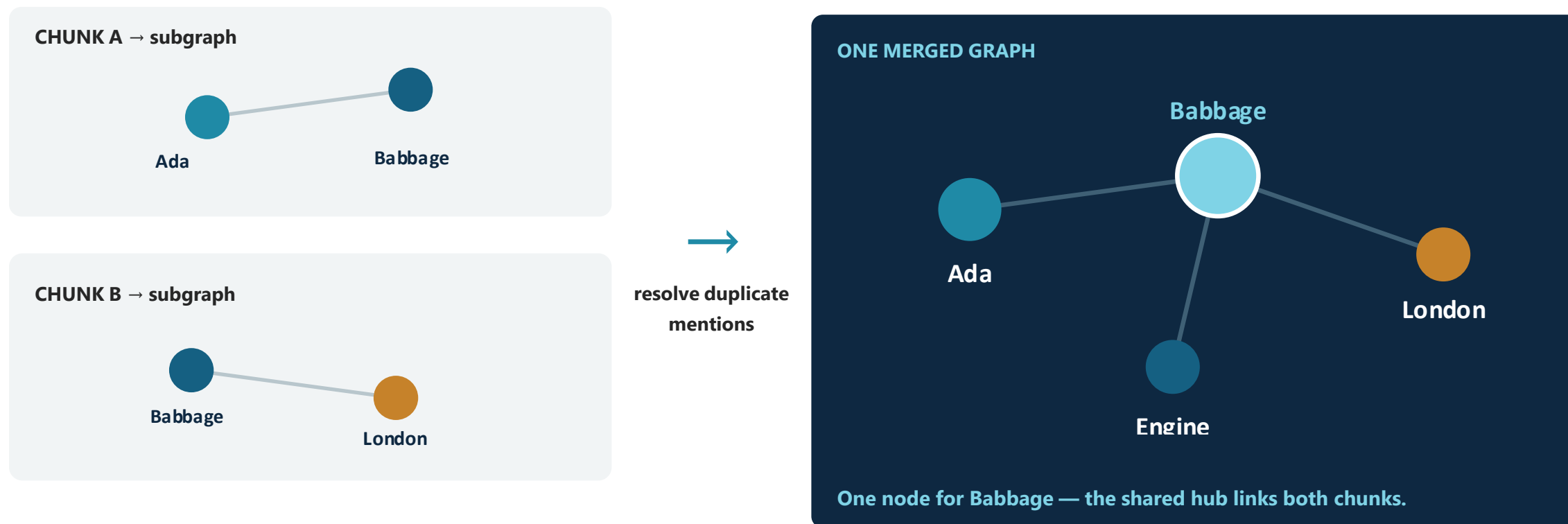
-----Community reports-----

Report 12: The Acme supply network. Acme Corp sits at the center of a cluster of suppliers and logistics partners...

-----Sources-----

[chunk 47] ...the agreement, signed in March 2023, commits Acme to exclusive component supply for a period of five years...

Step 3a: merge chunks into one graph



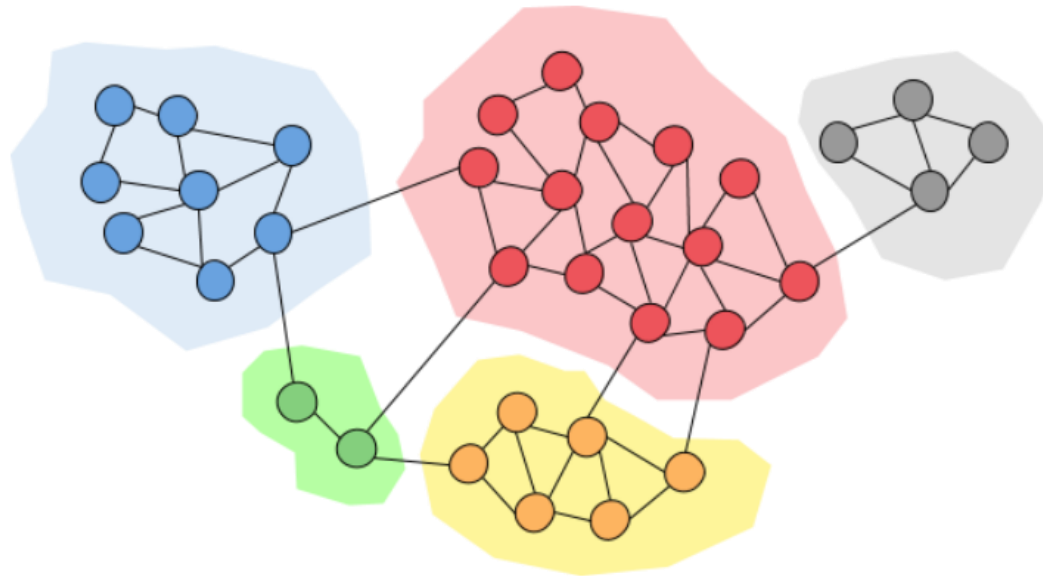
Entity resolution matches “Babbage / Charles Babbage / C. Babbage” **by name + embedding** → merges edges & descriptions onto one canonical node. This is what turns fragments into a connected corpus-wide graph.

Step 3: Community Detection

*Q: how do I **pre-organize this graph** so that a future question about "**the whole corpus**" is answerable without reading the whole corpus?*

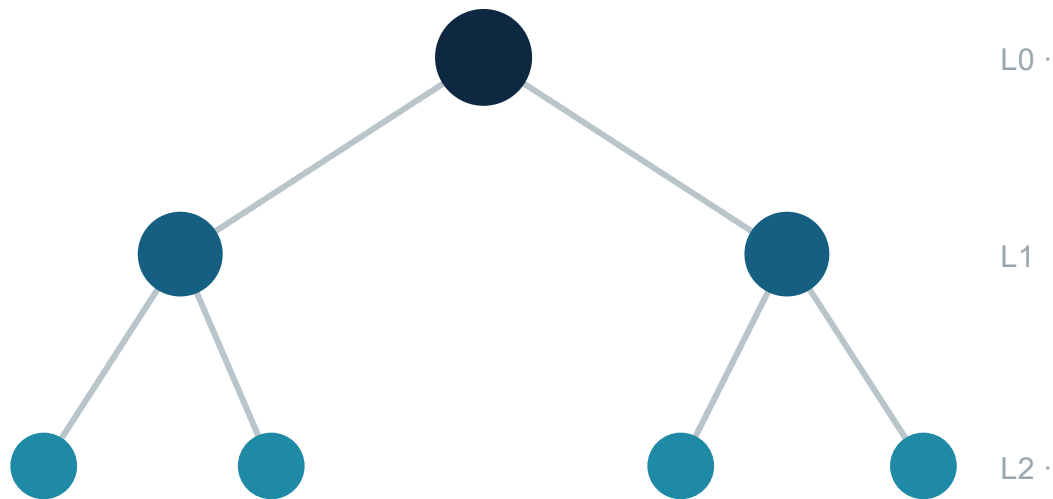
Partition the graph into clusters of related "entities" at multiple resolutions

Leiden Algorithm



Step 4: a summary per community, per level

LEIDEN HIERARCHY



COMMUNITY REPORT · L0

"Early computing pioneers and their machines — Lovelace, Babbage, and the Analytical Engine..."

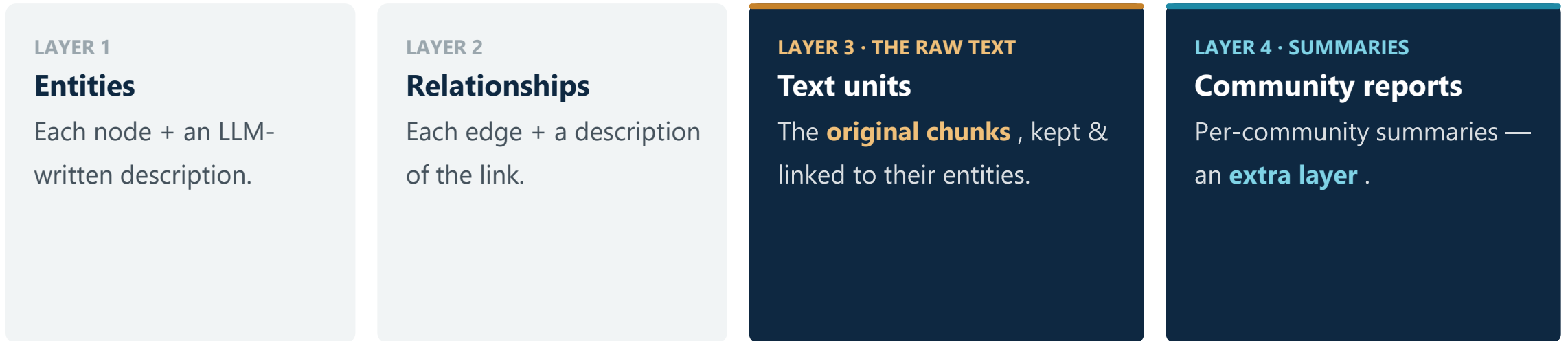
COMMUNITY REPORT · L2

"The Analytical Engine: design in London, its mill and store, and Lovelace's notes..."

Each report summarises a community's **entities, relations & claims** — written by the LLM.

Summaries or The Whole Text?

The index stores **four layers**. The original source text is never discarded — the summaries sit *on top* of it:



Which layer is used depends on the query → **local search** reads the source text; **global search** reads the summaries.

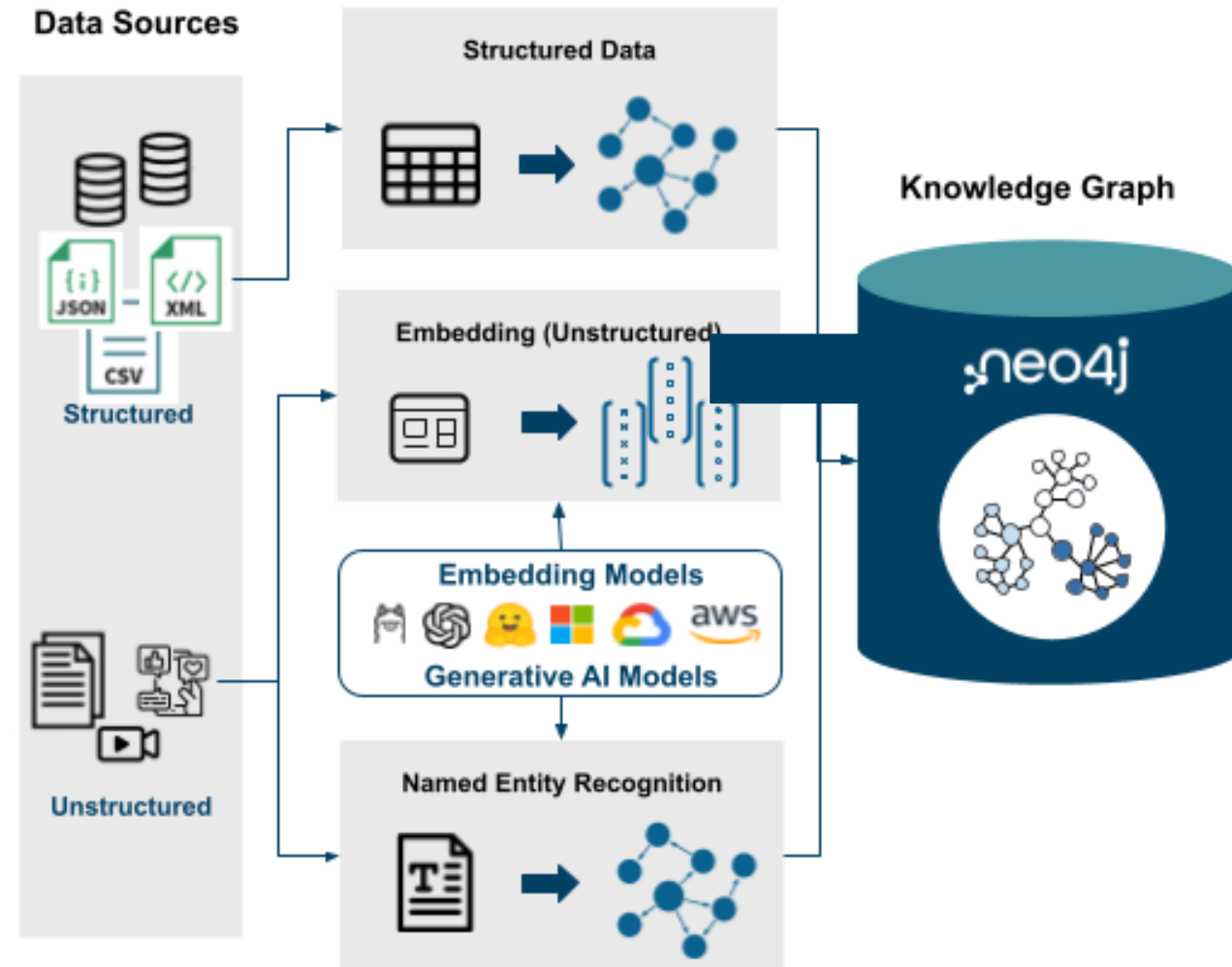
Query time – Local Search

"What did Babbage design, and where?"



Local search reads the original source text (+ entity descriptions & nearby summaries), reached through the graph — precise and traceable.

Knowledge Graph Builder



Benefits vs Risks of Traditional GraphRAG

Expensive Indexing

Graph is built by LLM prompting → Extracting entities and relationships with an LLM means **many API calls** over your whole corpus

Indexing Quality

Graph is built by LLM prompting → LLM that builds the graph may **hallucinate relationships** between elements in the corpus.

Entity Duplication: “NEOCHIP” and “NeoChip Inc.” result in two nodes.

Manual Query

Global vs Local option chosen manually at query time.

```
graphrag query --method global
```

“GRAPH” RAG?

Node retrieval is just vector search, but over entity descriptions instead of text chunks

SECTION 3

G-Retriever

The setup: a question over a huge graph

The **answer lives in a small, connected region of a graph** too big to feed the LLM whole. Two naïve options both fail:

Dump the whole graph

Doesn't fit the context window — and the LLM drowns in irrelevant nodes.

✗ too big

Top-k independent nodes

Like plain GraphRAG — returns disconnected fragments; the **connecting edges are missing**.

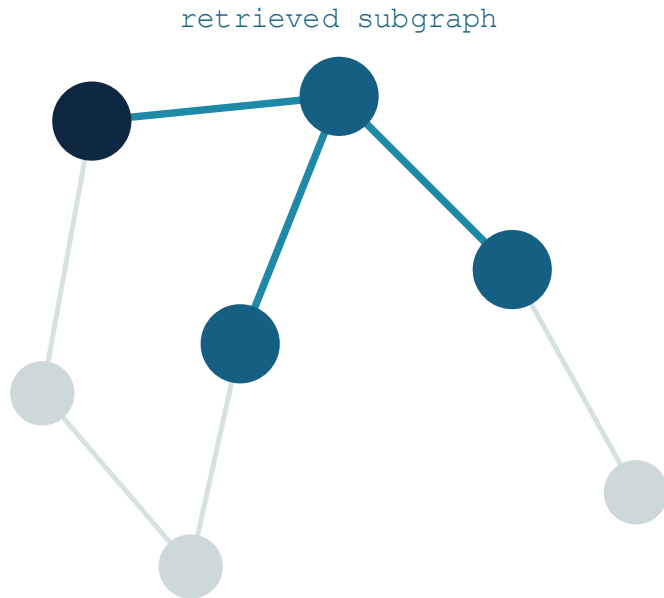
✗ chain broken

What we need

A **connected, relevant, right-sized subgraph** — the region that holds the answer.

✓ G-Retriever

Retrieve a Sub-graph, not chunks



PRIZE-COLLECTING STEINER TREE

$$\max \sum \text{prize}(\text{node}) - \sum \text{cost}(\text{edge})$$

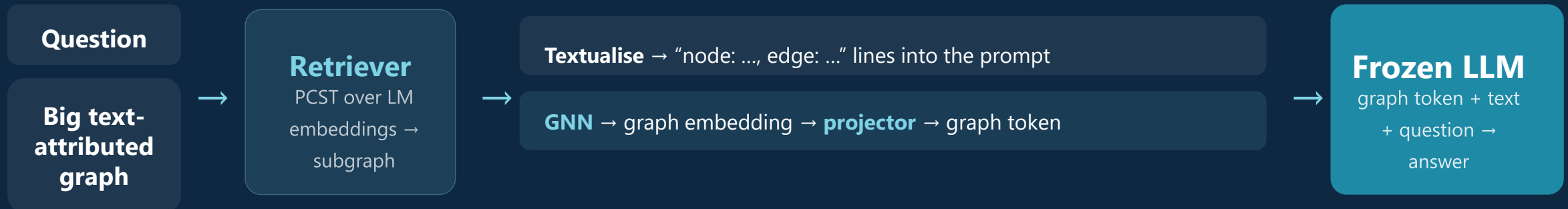
- **Prize** = relevance of a node/edge to the question
- **Edge cost** keeps the subgraph small & connected
- A principled retriever — not a top-k heuristic

G-Retriever

He et al. G-retriever: Retrieval-augmented generation for textual graph understanding and question answering. NeurIPS 2024.

- | | | | |
|---|-----------------------|--|-----------|
| 1 | Index | Encode every node & edge text with a frozen LM — once. | RETRIEVAL |
| 2 | Retrieve | Score nodes & edges by similarity to the question. | RETRIEVAL |
| 3 | Build subgraph | Run PCST → a connected, right-sized subgraph. | RETRIEVAL |
| 4 | Encode (GNN) | Message passing over the subgraph → one graph embedding. | REASONING |
| 5 | Generate (LLM) | Graph token + textualised subgraph → frozen LLM answers. | REASONING |

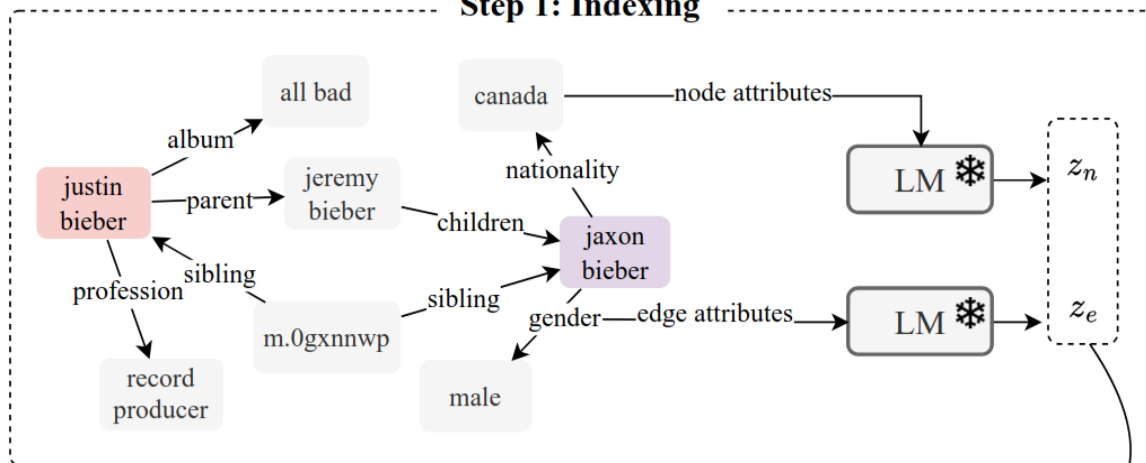
The G-Retriever Pipeline



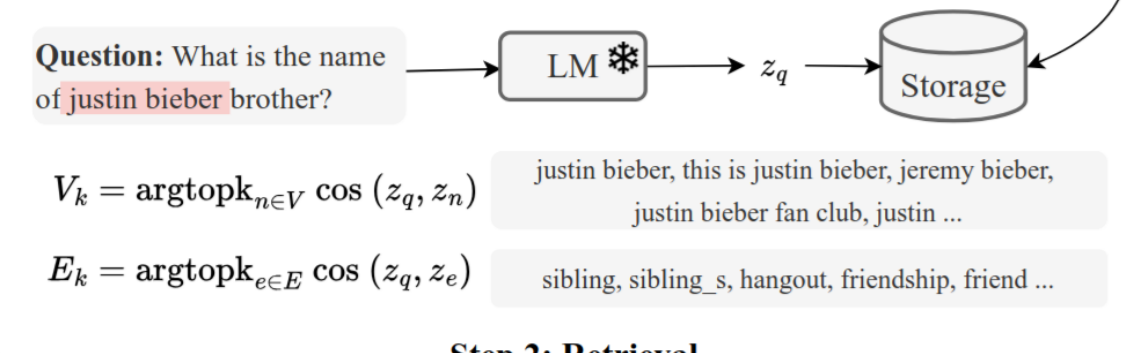
Two paths into one model: text the LLM can read + a learned token that carries the structure.

G-Retriever

Step 1: Indexing



Question: What is the name of justin bieber brother?



$$V_k = \text{argtopk}_{n \in V} \cos(z_q, z_n)$$

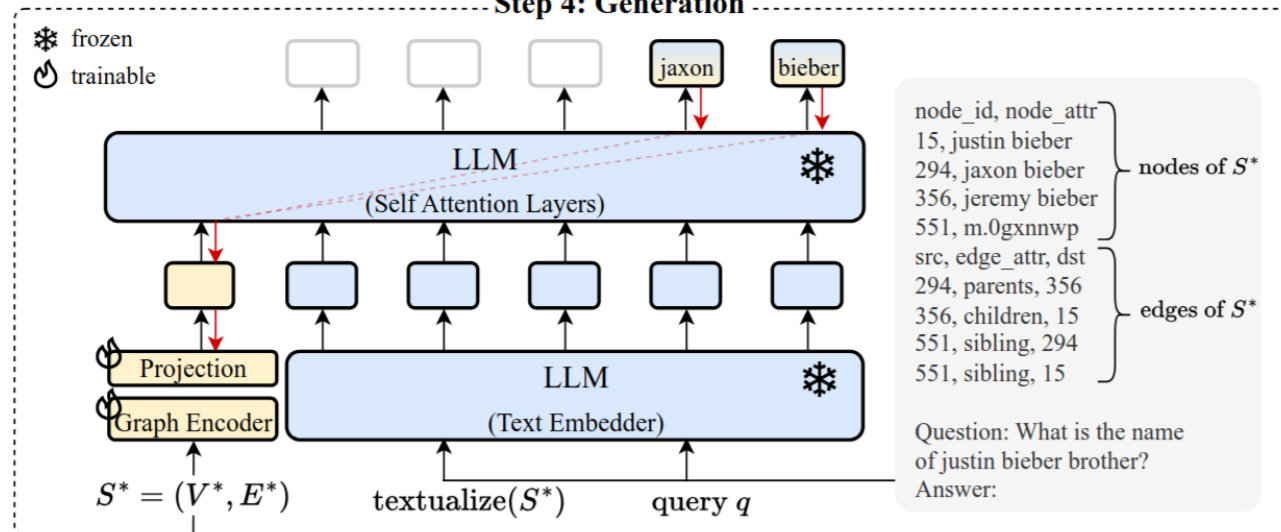
justin bieber, this is justin bieber, jeremy bieber, justin bieber fan club, justin ...

$$E_k = \text{argtopk}_{e \in E} \cos(z_q, z_e)$$

sibling, sibling_s, hangout, friendship, friend ...

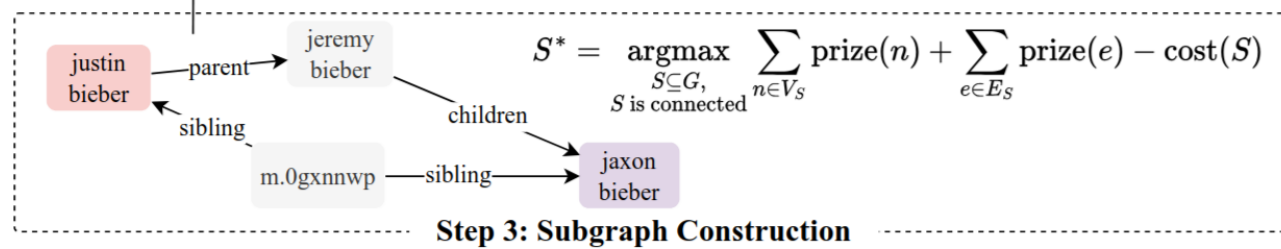
Step 2: Retrieval

Step 4: Generation



node_id, node_attr
15, justin bieber
294, jaxon bieber
356, jeremy bieber
551, m.0gxnnwp
src, edge_attr, dst
294, parents, 356
356, children, 15
551, sibling, 294
551, sibling, 15

Question: What is the name of justin bieber brother?
Answer:



Step 3: Subgraph Construction

$$S^* = \underset{S \subseteq G, S \text{ is connected}}{\text{argmax}} \sum_{n \in V_S} \text{prize}(n) + \sum_{e \in E_S} \text{prize}(e) - \text{cost}(S)$$

Image from paper. He X, Tian Y, Sun Y, Chawla NV, Laurent T, LeCun Y, Bresson X, Hooi B.

G-retriever: Retrieval-augmented generation for textual graph understanding and question answering. NeurIPS 2024.

Dr. Daniel F. Pérez-Ramírez | LLMs & GNNs for Reasoning Beyond Text

Practical Judgement

Worth the complexity when...

- Structure genuinely carries the answer
- Questions are multi-hop or relational
- The graph is reliable and maintained

Watch the costs

- **Latency** — retrieval + two model passes
- **Graph dependency** — stale graph poisons answers
- **Evaluation** — multi-hop correctness is hard to measure

If a plain LLM call or simple RAG already answers well — use that. Reach for the graph only when the structure is the key.

Lab preview: multi-hop QA on HotpotQA

HOTPOTQA — THE TASK

Paragraphs / question	10
-----------------------	----

Supporting (gold)	2
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Reasoning	multi-hop
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Cast as	node classification
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Yang et al., 2018

THE PIPELINE YOU BUILD

- ① Build a graph: paragraphs → **nodes**, shared entities → edges
- ② **GNN retriever** scores the supporting paragraphs
- ③ Build the subgraph of top nodes
- ④ Feed it to the **LLM reader** → final answer

Find 2 supporting paragraphs out of 10 — the GNN learns to spot them; the LLM alone misses them.